

CSCI 351 Python Programming Assignment – Checksums Part 1

In this assignment you will be processing packet headers and calculating checksum values. Your program will accept as input a single packet text file as input and then provide an organized output of the header fields, followed by the checksum calculations.

An file (input.txt) has been posted on mycourses. This file can be opened by Wireshark.

Your program must accomplish the following:

- Provide usage and/or operation messaging
- Process the Ethernet header fields
- Process the IP header fields
- Process the TCP header fields
- Calculate the IP and TCP header checksum and compare them to the actual field values
- Document sources used in this program

Note: You will be using the default installation of Python. I just upgraded to 3.14.7 so we will see how it goes. I will not install or build anything special for this assignment. You are not permitted to utilize specialized libraries though you may use anything included with Python 3. Examples: checksum, isnumeric, bin, etc.

To be clear – this is part 1. You must build everything from scratch. No libraries (ex. pyshark) or imports that are not your files.

Checksum video: <https://www.youtube.com/watch?v=7LniwyiH0SM>

IPv4 RFC: <https://www.rfc-editor.org/rfc/rfc791>

TCP RFC: <https://www.rfc-editor.org/rfc/rfc9293>

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Rubric

Name and contact info in code	/5
Python style guide (PEP-8) adherence	/5
Comments used for all functions/processes	/5
Operates per the specification	/10
Read from the file properly	/10
Exception catching (bad file, etc.)	/5
Neatly formats the output: field name and value	/35
Header checksums calculated per spec and correct	/20
Sources documented	/5
Bonus – Take one of the challenging problems in this program and ask your favorite generative AI tool a targeted question for the problem. Provide a screenshot and compare its result to your solution. Quality answers earn higher points.	
	/up to 10